

11	<u>Intended</u> - Brown paper package - Paper - Glue <u>Used</u> All	<u>Utensils</u> 1. Through a local vendor, source the necessary packages. Refer to Appendix XX to understand the reasons behind this decision. 2. Using a printer, print the created packaging of Snacktive based on Crit B Str 4. 3. Using commercial packaging glue, stick onto the acquired package. 4. Transfer 100g of Snacktive into the pack and seal for sale. <u>Used</u> All Under the time limit	
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Strand 4: Justify changes made to the plan

- No changes were made to the logical plan or the fundamental structure of the logical plan. The only notable change would be the skipping of Step 2 due to Expert Recommendation.

Change 1: Testing the oven using baked vegetables instead of salt crackers

Justification: This was a change made to the plan due to the convenience of using already diced vegetables in a seasoning mix instead of making a difficult dough in order to make salt crackers. These baked vegetables allowed me to understand how much time it took to cook them through, the likelihood of burning, and overall manage my expectations from the appliance. However, putting in the effort to make salt crackers would have allowed me to gauge the time required to create Snacktive, and understand the way dough behaves in the oven. Such a change to the logical plan is simple and without much consequence to the creation of the final product.

Change 2: Testing the blender by making Chimichurri and not Breadcrumbs

Justification: The goal of this step was to understand how the blender functions, the way the various speed settings interact with the contents inside the jar, and what jar is best suited to dry and wet ingredients. To fully test my knowledge on this particular topic, it made sense to make something that had components that were both wet and dry. Chimichurri was the perfect mix as garlic, dried chillies and other dried herbs required blending, eventually to be mixed with olive oil and blended cilantro - allowing for both dry and wet ingredients to be used. Using breadcrumbs to test my knowledge would have been without great consequence at all, only testing my knowledge on how to use dry blender jars and lower power settings.

Change 3: Testing the stove by making Chinese Style Stir-Fry and not Sauteed Vegetables or Steamed Beans

Justification: The goal of this step was to test my understanding of the stove, demonstrate compliance with safety protocol and create something that was cooked, not burnt and hopefully delicious. It became clear that in order to demonstrate all these skills, Chinese Stir-fry would have been most beneficial. The process of making this dish involves high-heat cooking, steaming with minimal water and deglazing with various sauces - a combination that would challenge my skills across the board.

Change 4: Changing the composition of Snacktive to improve texture

Justification: During initial tests of the Snacktive dough, the usage of boiled black beans and cooked quinoa caused the dough to become overly soggy, sticky and not capable of maintaining a shape. Even when baked, the texture was more crumbly than crisp - becoming similar to a cookie and not a chip or cracker. As the fundamental goal of this project is to create a viable alternative to traditional snacking options such as potato chips or other baked crisps - this texture would not be capable of meeting this goal. As a result, some change in the composition was necessary.

The understanding gained from these trials is that there was a need for significantly more starch in order to absorb moisture and become crisp in the oven. Approximately 4 tablespoons of extra oat flour was required, and in order to counteract the blandness caused by this, extra salt and cumin was needed. The ability to knead was significantly increased, adhering to a conformed shape and maintaining a consistent thickness. Additionally, when baked, it was able to retain its shape, become crisp, and have a cracker like texture.

Change 5: Adding Baking Powder (sodium bicarbonate) to improve texture.

Justification: Even with the success of the new dough composition, the resulting texture was flat and hard, almost like biting into a hard rock. This type of texture is not only not pleasant to any consumer but can also potentially become a hazard, resulting in micro cuts within the mouth. Adding sodium bicarbonate would help incorporate air into the dough, and make it fluffier - giving it a flaky and crispy texture instead of hard and flat. In the trial performed, this addition proved to be a step in the right direction - finally providing a baked crisp which was crunchy, flaky and dry.

Change 6: Choosing not to test the use of a blender

Justification: After an in-depth conversation with the expert and analysis conducted regarding the various processes that involve the processing of foods, it can be concluded that the use of a blender itself is not needed. This is particularly true due to the fact that a simple masher, or a humble fork can be used to reach the same point without potentially creating a hazardous environment in the cooking space. It was more efficient to use a fork to mash the baked beans into the quinoa than to blend them together, which would require large blender jars and involve an extensive clean-up process.

Criterion D: Evaluating

Strand 1: Design Testing Methods

Testing Method	Process	Purpose	Specifications Tested
Nutritional Analysis (<i>Quantitative Analysis of Snacktive's ability to provide nutrition</i>)	Basic nutritional analysis will help determine the basic nutritional quantities like energy, carbohydrates, proteins, fats, sugars, etc. This involves contacting an established food testing lab and conducting thorough nutritional analysis. (Refer Appendix A5)	Understanding the values found in Snacktive will help evaluate its ability to provide nutrition, and assess the need to make improvements to the design. Professional nutritional analysis helps do this.	Nutrition
Focus Group Survey (<i>Qualitative analysis of the packaging</i>)	A focus group is first determined through identifying the various stakeholders involved. This group would first be shown the packaging design, and would then provide their feedback on it. Following which, the lead individual would inquire on the lines of <i>does this packaging echo sustainability? Would you buy a snack that comes in this box? What looks good? What doesn't look good? Is the</i>	Responses by the stakeholders helps gain a deeper understanding of their thought process. This increased understanding helps evaluate the success of the solution, paving the way for necessary changes being implemented.	Packaging Aesthetics Cost